

Classifier Reliability Estimation from Pseudo-Label Agreement: EM and Variational Bayes Inference

1 Methodology

1.1 Problem Formulation

Let $\mathcal{M} = \{M_j\}_{j=1}^J$ denote a pool of classifiers. For each instance x_i , classifier j produces prediction

$$r_i^j = M_j(x_i).$$

A pseudo-label $\hat{y}_i$ is available but may be noisy.

We define the observed agreement variable

$$C_i^j = \mathbf{1}(r_i^j = \hat{y}_i),$$

which indicates whether classifier j agrees with the pseudo-label.

1.2 Latent Correctness Variables

Let

$$Z_i^j = \mathbf{1}(r_i^j = y_i),$$

where y_i is the unknown true label.

Hence,

$$Z_i^j = \begin{cases} 1, & \text{if classifier } j \text{ predicts the true label correctly,} \\ 0, & \text{otherwise.} \end{cases}$$

Each classifier has accuracy parameter

$$\alpha_j = P(Z_i^j = 1).$$

The pseudo-label quality is characterized by

$$\beta = P(\hat{y}_i = y_i).$$

1.3 Generative Model

For each classifier and instance,

$$Z_i^j \mid \alpha_j \sim \text{Bernoulli}(\alpha_j).$$

Conditional on Z_i^j ,

$$C_i^j \mid Z_i^j, \beta \sim \text{Bernoulli}(\theta_i^j),$$

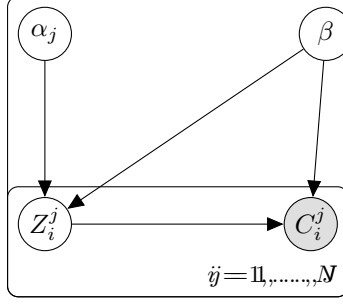


Figure 1: Probabilistic graphical model for classifier reliability estimation. The latent variable Z_i^j indicates whether classifier j predicts the true label correctly on instance i . The observed variable C_i^j records agreement between the classifier prediction and the pseudo-label.

where

$$\theta_i^j = \begin{cases} \beta, & Z_i^j = 1, \\ 1 - \beta, & Z_i^j = 0. \end{cases}$$

Equivalently,

$$P(C_i^j = 1 \mid Z_i^j, \beta) = \beta^{Z_i^j} (1 - \beta)^{1 - Z_i^j}.$$

1.4 Model Intuition

The hidden variable Z_i^j represents the correctness of classifier j on instance i . Since the true label is unavailable, classifier correctness cannot be observed directly. Instead, agreement with the pseudo-label, represented by C_i^j , provides indirect evidence regarding correctness. When pseudo-labels are reliable, correct classifiers are more likely to agree with them than incorrect classifiers.

Therefore, agreement observations act as noisy measurements of latent classifier correctness. The parameters α_j capture classifier accuracies, while β measures global pseudo-label quality.

2 Complete-Data Likelihood

The complete-data likelihood is

$$p(\mathbf{C}, \mathbf{Z} \mid \alpha, \beta) = \prod_{i=1}^N \prod_{j=1}^J p(Z_i^j \mid \alpha_j) p(C_i^j \mid Z_i^j, \beta).$$

Its logarithm is

$$\log p(\mathbf{C}, \mathbf{Z} \mid \alpha, \beta) = \sum_{i=1}^N \sum_{j=1}^J \left[Z_i^j \log \alpha_j + (1 - Z_i^j) \log(1 - \alpha_j) \right] \quad (1)$$

$$+ \sum_{i=1}^N \sum_{j=1}^J \left[Z_i^j (C_i^j \log \beta + (1 - C_i^j) \log(1 - \beta)) \right. \quad (2)$$

$$\left. + (1 - Z_i^j) (C_i^j \log(1 - \beta) + (1 - C_i^j) \log \beta) \right]. \quad (3)$$

3 Expectation-Maximization

3.1 General EM

Given latent variables Z , observations X , and parameters θ , EM iteratively maximizes

$$\log p(X \mid \theta).$$

E-step:

$$q(Z) = p(Z \mid X, \theta^{(t)}).$$

M-step:

$$\theta^{(t+1)} = \arg \max_{\theta} \mathbb{E}_q[\log p(X, Z \mid \theta)].$$

3.2 E-Step

Define

$$\tau_i^j = P(Z_i^j = 1 \mid C_i^j, \alpha_j, \beta).$$

$$\tau_i^j = \frac{p(Z_i^j = 1 \mid \alpha_j)P(C_i^j \mid Z_i^j, \beta)}{p(Z_i^j = 1 \mid \alpha_j)P(C_i^j \mid Z_i^j, \beta) + p(Z_i^j = 0 \mid \alpha_j)P(C_i^j \mid Z_i^j, \beta)} \quad (4)$$

Using Bayes' rule,

$$\tau_i^j = \frac{\alpha_j \beta^{C_i^j} (1 - \beta)^{1 - C_i^j}}{\alpha_j \beta^{C_i^j} (1 - \beta)^{1 - C_i^j} + (1 - \alpha_j) (1 - \beta)^{C_i^j} \beta^{1 - C_i^j}}.$$

Equivalently,

$$\tau_i^j = \sigma \left(\log \frac{\alpha_j}{1 - \alpha_j} + (2C_i^j - 1) \log \frac{\beta}{1 - \beta} \right).$$

3.3 M-Step

Updating classifier accuracies gives

$$\alpha_j^{new} = \frac{1}{N} \sum_{i=1}^N \tau_i^j.$$

Updating pseudo-label quality gives

$$\beta^{new} = \frac{1}{NJ} \sum_{i=1}^N \sum_{j=1}^J \left[\tau_i^j C_i^j + (1 - \tau_i^j) (1 - C_i^j) \right].$$

4 Variational Bayes Inference

Assume Beta priors

$$\alpha_j \sim \text{Beta}(a_0, b_0), \quad \beta \sim \text{Beta}(c_0, d_0).$$

We use the mean-field approximation

$$q(\mathbf{Z}, \alpha, \beta) = q(\beta) \prod_{j=1}^J q(\alpha_j) \prod_{i=1}^N \prod_{j=1}^J q(Z_i^j).$$

Let

$$\begin{aligned} q(Z_i^j) &= \text{Bernoulli}(\phi_i^j), \\ q(\alpha_j) &= \text{Beta}(\tilde{a}_j, \tilde{b}_j), \\ q(\beta) &= \text{Beta}(\tilde{c}, \tilde{d}). \end{aligned}$$

4.1 Latent Variable Updates

$$\phi_i^j = \sigma(\Lambda_i^j),$$

where

$$\Lambda_i^j = (E[\log \alpha_j] - E[\log(1 - \alpha_j)]) + (2C_i^j - 1)(E[\log \beta] - E[\log(1 - \beta)]).$$

4.2 Updates for $q(\alpha_j)$

$$\begin{aligned} \tilde{a}_j &= a_0 + \sum_{i=1}^N \phi_i^j, \\ \tilde{b}_j &= b_0 + \sum_{i=1}^N (1 - \phi_i^j). \end{aligned}$$

4.3 Updates for $q(\beta)$

$$\begin{aligned} \tilde{c} &= c_0 + \sum_{i,j} \left[\phi_i^j C_i^j + (1 - \phi_i^j)(1 - C_i^j) \right], \\ \tilde{d} &= d_0 + \sum_{i,j} \left[\phi_i^j (1 - C_i^j) + (1 - \phi_i^j) C_i^j \right]. \end{aligned}$$

4.4 Required Expectations

$$\begin{aligned} E[\log \alpha_j] &= \psi(\tilde{a}_j) - \psi(\tilde{a}_j + \tilde{b}_j), \\ E[\log(1 - \alpha_j)] &= \psi(\tilde{b}_j) - \psi(\tilde{a}_j + \tilde{b}_j), \\ E[\log \beta] &= \psi(\tilde{c}) - \psi(\tilde{c} + \tilde{d}), \\ E[\log(1 - \beta)] &= \psi(\tilde{d}) - \psi(\tilde{c} + \tilde{d}). \end{aligned}$$

5 Algorithms

EM alternates between computing posterior correctness probabilities τ_i^j and updating parameters α_j and β .

VB alternates between updating the variational posterior probabilities ϕ_i^j and the Beta posterior parameters $(\tilde{a}_j, \tilde{b}_j)$ and $(\tilde{c}, \tilde{d})$ until convergence of the ELBO.